



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
SOMETHINGANCHAL CAMPUS**

**A MAJOR PROJECT PROGRESS REPORT ON
IMPLEMENTING A SUPER COOL PROJECT USING ALL THE
KNOWLEDGE I HAVE ACQUIRED.**

BY

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MARCH, 2026

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A project submitted to the Department of Something and Something Engineering in partial
fulfillment of the requirements for the Bachelor's Degree in Computer Engineering

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March, 2026

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ABSTRACT

Your Abstract

Keywords: *filesystem, microkernel, copy-on-write, scheme, system calls*

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LIST OF ABBREVIATIONS

BSD	Berkeley Software Distribution
CD	Compact Disc
COW	Copy On Write
FAT	File Allocation Table
I/O	Input/Output
IPC	Inter-Process Communication
OS	Operating System
ROM	Read-Only Memory
RAM	Random Access Memory
USB	Universal Serial Bus
VM	Virtual Machine

CHAPTER 1

INTRODUCTION

1.1 Background

Background

1.2 Gap Identification

Gaps

1.3 Motivation

Motiv

1.4 Objectives

Objectives

- First Objective
- Second Objective

CHAPTER 2

RELATED THEORY

2.1 Something

2.1.1 Introduction

Intro

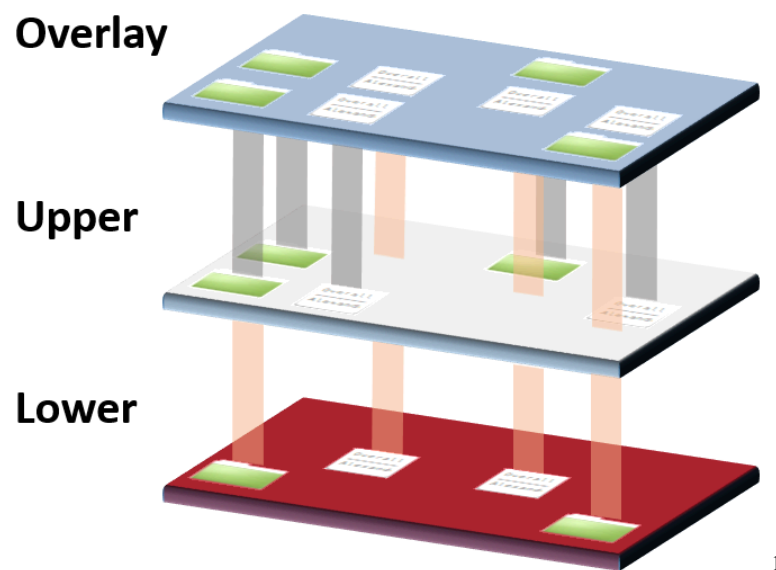


Figure 1: Visual Representation of Layers in an Overlay Filesystem

2.2 Design

Something Else

¹E. C. Design, “Understand What an OverlayFS Is and How It Works.” [Online]. Available: <https://embeddedcomputing.com/technology/processing/understand-what-an-overlayfs-is-and-how-it-works>

CHAPTER 3

LITERATURE REVIEW

In virtualization, OverlayFS offers an efficient method for managing virtual machine disk images [1].

CHAPTER 4

METHODOLOGY

Detailed explanation of methodology used.

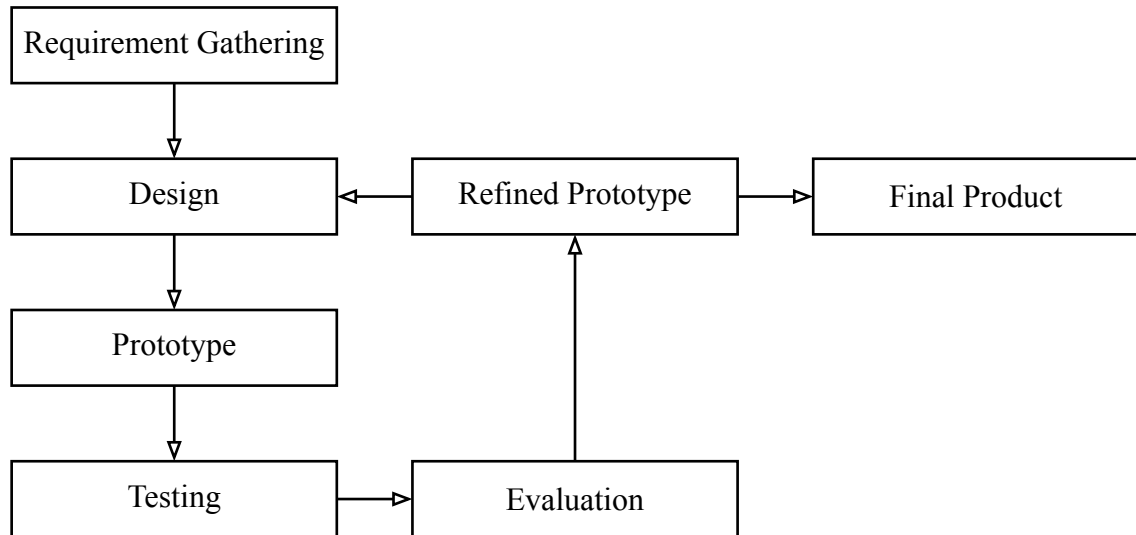


Figure 2: Prototype Model

4.1 Development Approach

Iterative Approach / The approach you used.

CHAPTER 5

IMPLEMENTATION

5.1 Architecture

How it functions.

5.2 Something Else

Something

CHAPTER 6

RESULTS / CURRENT STATUS

Results

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

This is what we achieved.

7.2 Future Work

It can be further improved.

REFERENCES

- [1] J. Scholz, “Using OverlayFS to Save Disk Space.” [Online]. Available: <https://e2b.dev/blog/scaling-firecracker-using-overlayfs-to-save-disk-space>

APPENDIX

Project Timeline

The following Gantt chart illustrates our planned project timeline from initial research through final presentation. The timeline is divided into three major phases: Research Phase, Development Phase, and Final Phase.

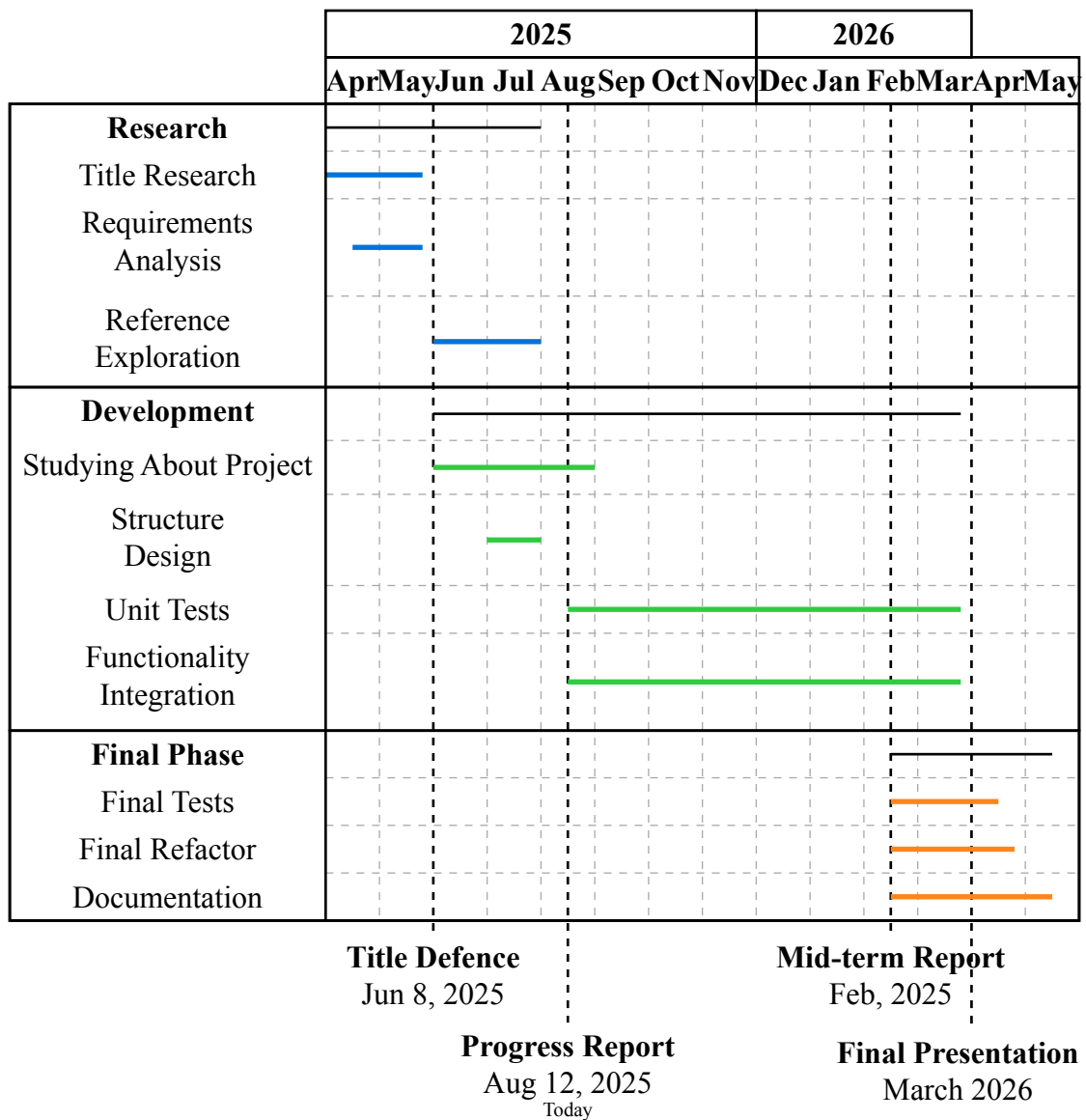


Figure 3: Project Timeline